higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

PROCESS CHEMISTRY
NQF LEVEL 4

(6050444)

13 November 2019 (X-Paper)
09:00–12:00

This question paper consists of 6 pages.

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TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Read ALL the questions carefully.
2. Answer ALL the questions.
3. Number the answers according to the numbering system used in this question paper.
4. Start each question on a NEW page.
5. Write neatly and legibly.

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QUESTION 1

Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (1.1–1.5) in the ANSWER BOOK.

1.1 When heated, thermo-hardening polymers undergo a reversible physical change.

1.2 Platinum is used in the jewellery industry.

1.3 Asphalt is a by-product from petroleum and can be used for gas heaters.

1.4 Butane can be used as a fuel in cigarette lighters.

1.5 Packed papers should be stored in a warehouse that is dry.

(5 × 1) **[5]**

QUESTION 2

Choose an item from COLUMN B that matches a description in COLUMN A. Write only the letter (A–H) next to the question number (2.1–2.5) in the ANSWER BOOK.

COLUMN A		COLUMN B	
2.1	Cross-linked polyethylene	A	hydroxybenzene
2.2	A mass of liquid bubbles	B	cumene
2.3	Common name for isopropyl benzene	C	antibiotic
2.4	Another name for phenol	D	PEX
2.5	Pharmaceuticals used to kill or stop the growth of disease-causing bacteria	E	froth
		F	cyclohexane
		G	probiotics
		H	LDPE

(5 × 1) **[5]**

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QUESTION 3

Give ONE word or a term for each of the following descriptions. Write only the word or term next to the question number (3.1–3.5) in the ANSWER BOOK.

3.1 A substance that accelerates a chemical reaction without itself being modified.

3.2 A process used to increase the concentration of the desired base metal by air sparging through slurry.

3.3 Any chemical process of converting a substance into a gas by high temperature heating.

3.4 A temperature at which a liquid solidifies under isobaric conditions.

3.5 An organism that sporulates during unfavourable conditions.

(5 × 1) **[5]**

QUESTION 4

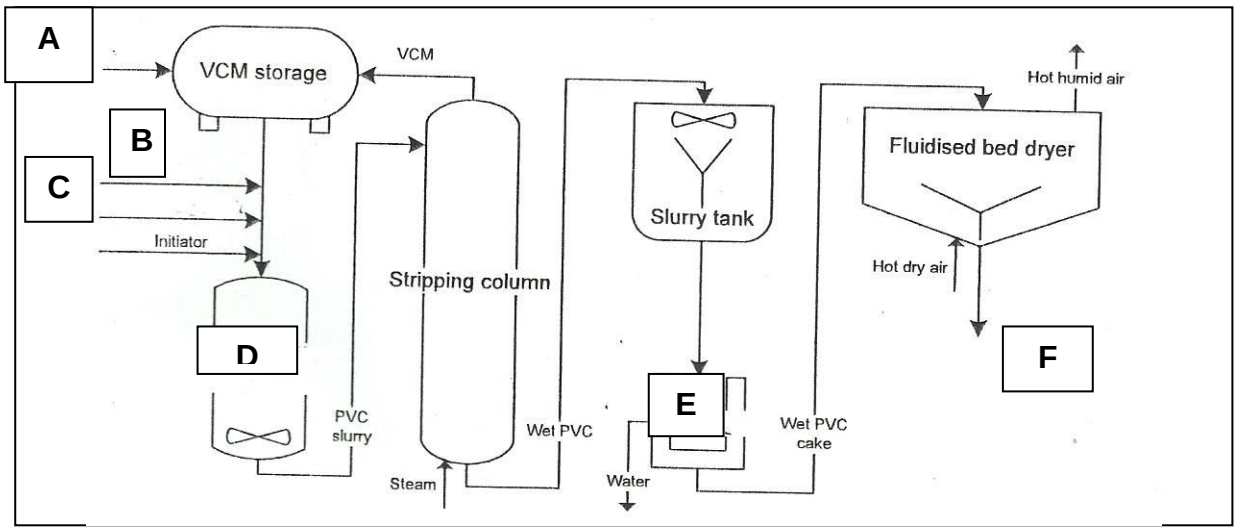
4.1 Lubricants are liquids or powdered organic compounds which are added in small quantities during polymer processing.

4.1.1 Name TWO types of lubricants. (2)

4.1.2 State THREE important functions of lubricants in polymer processing. (3)

4.2 The following diagram is a process flow diagram of a polyvinyl chloride (PVC) suspension polymerisation process.

Complete the diagram by filling in the missing words. Write the letter (A–F) next to the answer in the ANSWER BOOK.



(6)

4.3 Use the diagram in QUESTION 4.2 above to answer the following:

4.3.1 Name the process used to remove VCM from the stripping columns to VCM storage tanks. (1)

4.3.2 What is the function of part E? (1)

4.3.3 What is the primary function of an initiator? (1)

4.3.4 What agent is used to dry PVC? (1)

4.4 Name the THREE elemental constituents (species of atoms) of polyvinyl chloride. (3)

4.5 List FOUR categories of polymers. (4)

4.6 State FIVE methods used during plastic conversion. (5)
[27]

QUESTION 5

5.1 Palladium is one of the precious metals produced in South Africa.

List FOUR uses of palladium. (4)

5.2 Sketch a labelled flow diagram for ferroalloy production. (13)

5.3 South Africa is the largest producer of ferrovanadium.

List FIVE locations where you can find vanadium containing magnetite ores of economic importance. (5)
[22]

QUESTION 6

6.1 Xylene is a highly flammable liquid with a sweet smell.

6.1.1 What is the IUPAC name for xylene? (1)

6.1.2 List THREE isomers of xylene. (3)

6.1.3 Give TWO functions (applications) of xylene. (2)

6.2 List FIVE products obtained from the coking process. (5)

6.3 Briefly describe how phenols are stripped from a gas. (5)

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6.4 Kerosene is widely used in the UK and Southeast Asia.

6.4.1 What is another name for kerosene? (1)

6.4.2 State THREE uses of kerosene. (3)

6.5 List THREE processes used to refine crude oil. (3)
[23]

QUESTION 7

7.1 List FOUR primary raw materials that are used in beer brewing. (4)

7.2 Discuss the steeping process in beer brewing. (4)

7.3 Sequentially, name THREE seed (inoculum) preparation steps used in the production of penicillin. (3)

7.4 Briefly describe cooling and its purpose in downstream processing of crude penicillin broth. (2)
[13]

TOTAL: 100

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